LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Interrogation mode: - various interrogation and their pulse format
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Interrogation mode: - various interrogation and their pulse format.
5. Specific Objectives	18.1.3.1 Trainee will be able to describe the basic theory of operation of surveillance system of an aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of interrogation pulse formats using IB diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft surveillance system?
(b) How does the transponder respond to an interrogation signal?
(c) What are the two main types of interrogation modes used in air traffic control?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic theory of surveillance system operation • Mode 3/A interrogation pulse format • Mode 4/S interrogation pulse format • Pulse repetition frequency variations • Interrogation frequency bands	L & Ds	PPT, IB	15 min
(b)	• Comparison of Mode 3/A and Mode 4/S pulse widths • Identification of pulse spacing in interrogation signals • Role of guard intervals in pulse formats • Differences in reply pulse structures • Impact of pulse format on data transmission	L & Ds	PPT, PDF	20 min
(c)	• Analysis of pulse timing diagrams • Verification of pulse format standards • Troubleshooting common interrogation errors • Review of pulse synchronization techniques • Summary of key pulse format characteristics	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between Mode 3/A and Mode 4/S pulse formats.
(b) How does the pulse repetition frequency affect system performance?
(c) Describe the role of guard intervals in interrogation signals.

12. Home Assignment and Follow Up	Prepare a brief note on the pulse format differences between Mode 3/A and Mode 4/S for the next lesson on emergency locator transmitters.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to emergency locater Transmitter. Basic principle of ELT.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min